

OUSSAMA GAIED

Electrical Engineer | Embedded Control and Robotics

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RESEARCH PROFILE

Electrical engineer (ENIM, 2025) who designs embedded control systems from algorithm to hardware. Mitacs Globalink research intern at Université de Moncton: implemented real-time model predictive control (MPC) on a distributed multi-node ESP32 system with environmental sensing, achieving a validated reduction of about 20% in heating energy consumption. Experience in PID and motor control, sensor integration and real-time embedded programming (C/C++, Python, STM32). Seeking a master's or PhD in robotics and electrical and electronics engineering, with interest in autonomous systems, embedded control and intelligent robotic platforms.

EDUCATION

National Engineering School of Monastir (ENIM), Tunisia 09/2022 to 09/2025

Master of Engineering (Diplôme National d'Ingénieur) in Electrical Engineering, Distinction

- Final-year project completed through the Mitacs Globalink program

Higher Institute of Technological Studies of Mahdia (ISET), Tunisia 09/2019 to 06/2022

Bachelor's degree (Licence nationale) in Electrical Engineering, Distinction

RESEARCH EXPERIENCE

Université de Moncton, Faculty of Engineering, Canada 05/2025 to 08/2025

Research Intern, Mitacs Globalink | Supervisor: Prof. Mohsen Ghribi

Topic: minimizing the energy consumption of electric heating

- Implemented energy optimization algorithms based on model predictive control (MPC) and presence detection, with a validated reduction of about 20% in heating energy consumption.
- Designed a real-time distributed control system on ESP32 with a Nextion HMI, a multi-node Wi-Fi/BLE architecture and BME680 environmental sensors.
- Designed four custom electronic boards in Fusion 360, including a power board; all fabricated and tested with 100% validated operation.
- Validated the complete system experimentally on a prototype and wrote a research report presenting the results.

TECHNICAL SKILLS

Control: Model predictive control (MPC), PID regulation, servo and stepper motor control

Programming: C, C++, Python, SCL

Embedded systems and sensing: ESP32, STM32, FPGA, CAN, Wi-Fi/BLE, RS232, BME680 sensors, Nextion, TouchGFX

Electronics: Altium Designer, Fusion 360, LTspice

Industrial automation: Siemens S7-1200 and S7-300, TIA Portal, Step 7, GRAFCET/GEMMA, WinCC SCADA, EPLAN

Other tools: LabVIEW, Excel, MATLAB

ENGINEERING EXPERIENCE

Actia Engineering Services, ENIM Fablab, Tunisia 07/2024 to 09/2024

Engineering Intern

- Designed a custom PCB for the CAN data-transmission stage of an autonomous shuttle, ensuring reliable communication between embedded systems.

Sancellia, Monastir, Tunisia 03/2022 to 06/2022

Bachelor's final project: automation of a toilet-paper packaging machine

- Analyzed the machine and designed new electrical schematics for its control cabinet; programmed a S7-1200 PLC.

SELECTED PROJECTS

- **Autonomous line-following robot:** STM32 microcontroller with a PID control algorithm written in C.
- **Electric motor control:** servo motor and stepper motor control on STM32 in C/C++ with a TouchGFX embedded interface.
- **Wagon-washing machine:** automation with Siemens S7-1200, WinCC SCADA and EPLAN.

LANGUAGES

Arabic: native | French: C1, advanced | English: B2/C1, professional